

Code:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <string.h>
5 #include <sys/wait.h>
6 #include <sys/time.h>
7 #define BUFFER_SIZE 1024
8 #define NUM_MESSAGES 10000
9 int main()
10 {
11     int fd[2];
12     pid_t pid;
13
14     char buffer[BUFFER_SIZE];
15     char message[BUFFER_SIZE];
16
17     struct timeval start, end;
18     double time_taken;
19
20     // Create unnamed pipe
21     if (pipe(fd) == -1)
22     {
23         perror("pipe");
24         exit(EXIT_FAILURE);
25     }
26
27     // Create child process
28     pid = fork();
29
30     if (pid < 0)
31     {
32         perror("fork");
33         exit(EXIT_FAILURE);
34     }
35
36     // Parent - Producer
37     if (pid > 0)
38     {
39         close(fd[0]); // Close reading end
40
41         gettimeofday(&start, NULL);
42
43         for (int i = 0; i < NUM_MESSAGES; i++)
44         {
45             sprintf(message, "Message %d from Producer", i + 1);
46             if (write(fd[1], message, strlen(message) + 1) == -1)
47             {
48                 perror("write");
49                 exit(EXIT_FAILURE);
50             }
51
52             close(fd[1]); // Close writing end
53
54             wait(NULL);
55
56             gettimeofday(&end, NULL);
57
58             time_taken =
59                 (end.tv_sec - start.tv_sec) +
60                 (end.tv_usec - start.tv_usec) / 1000000.0;
61
62             printf("\n--- Communication Efficiency ---\n");
63             printf("Number of messages : %d\n", NUM_MESSAGES);
64             printf("Time taken          : %.6f seconds\n", time_taken);
65
66             if (time_taken > 0)
67             {
68                 printf("Messages/second    : %.2f\n",
69                     NUM_MESSAGES / time_taken);
70             }
71
72             // Child - Consumer
73             else
74             {
75                 close(fd[1]); // Close writing end
76
77                 while (read(fd[0], buffer, BUFFER_SIZE) > 0)
78                 {
79                     // Consume the data
80                     // Uncomment the following line to display messages
81                     // printf("Consumer received: %s\n", buffer);
82                 }
83
84                 close(fd[0]);
85                 exit(EXIT_SUCCESS);
86             }
87
88             return 0;
89 }
}
```

Output:

```
--- Communication Efficiency ---
Number of messages : 10000
Time taken          : 0.007233 seconds
Messages/second     : 1382552.19
```